If we take quantum gravity into account, this prediction is no longer correct – there is quantum repulsion – the same repulsion that makes the universe bounce at the Big Bang. What we expect is that, on getting closer to the centre, the falling matter is slowed down by this quantum pressure, up to a very high but finite density. Matter gets squashed, but not all the way to an infinitely small point, because there is a limit to how small things can be. Quantum gravity generates a huge pressure that makes matter bounce out, precisely as a collapsing universe can bounce out into an expanding universe.

The bounce of a collapsing star can be very fast, if watched from down there. But – remember – time passes much more slowly there than outside. Seen from the outside, the process of the bounce can take billions of years. After this time, we can see the black hole explode. In the end, basically, this is what a black hole is: a shortcut to the distant future.

Thus, quantum gravity might imply that black holes are not eternally stable objects, as classical general relativity predicted, after all. They are, ultimately, unstable.

Seeing these black-hole explosions would be a spectacular confirmation for the theory. Very old black holes, such as those formed in the early universe, could be exploding today. Some recent calculations suggest that the signals of their explosion could be in the range of radio telescopes. It has even been suggested that certain mysterious radio pulses which radio astronomers have *already* measured, called Fast Radio Bursts, could be, precisely, signals generated by the explosion of primordial black holes. If this was confirmed, it would be fantastic: we would have a direct sign of a quantum gravitational phenomenon. Let’s wait and see …